

Sri Sivasubramaniya Nadar College of Engineering, Chennai

(An autonomous Institution affiliated to Anna University)

Department of Computer Science and Engineering

CAT Assignment 1

ICS1502 – Introduction to Machine Learning

Degree & Branch	M. Tech (Integrated) Computer Science & Engineering
Semester	V
Subject Code & Name	ICS1502 & Introduction to Machine Learning
Academic Year	2026–2027 (Odd)
Batch	2024–2029
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GitHub Repository	https://github.com/Amsavarthini25/ML_LAB_2470012
Due Date	28-08-2025

1 Objectives

- Construct the design matrix and label vector for the Mobile-Price data set and solve the ordinary-least-squares problem both analytically and iteratively.
- Quantify the impact of ℓ_2 regularization and standardization on prediction accuracy and on the relative importance of each feature.
- Train a logistic-regression model on the Bank-Note Authentication data, examine accuracy versus regularization strength, visualize the three most influential features in 3-D, and study the degradation caused by synthetic outliers.

2 Part 1.1 – Linear Regression (Mobile Phone Price Prediction)

2.1 Dataset Description

Dataset Name	Mobile Phone Price Prediction
Dataset Source	Local CSV (<code>Mobile-Price-Prediction.csv</code>); features describe hardware specifications of mobile phones.
Number of Samples	807 phones
Number of Features	7 continuous predictors: Ratings, RAM, ROM, Mobile_Size, Primary_Cam, Selfi_Cam, Battery_Power
Target Variable	Price (continuous, in INR)
Missing Values	Present in a few numerical columns; imputed by column-wise median before modelling.
Train-Test Split	80:20 (<code>random_state = 42</code>) $\Rightarrow$ 645 training samples / 162 test samples.
Pre-processing	Bias column of ones appended $\rightarrow$ design matrix $X \in \mathbb{R}^{645 \times 8}$. For gradient descent the features were standardized to zero mean and unit variance.

The design matrix X contains a leading column of ones so that the intercept (bias) is learned as the first component of the parameter vector $\theta \in \mathbb{R}^8$. The label vector $y \in \mathbb{R}^{645}$ holds the corresponding prices. This matrix formulation permits both the closed-form normal equation and gradient-descent updates to be written compactly.

2.2 1. Closed-form Solution (Ordinary Least Squares)

The normal equation without regularization is

$$\theta = (X^\top X)^{-1} X^\top y.$$

Obtained coefficients (first eight entries):

$$\theta_{\text{CF}} \approx [-1.034 \times 10^5, 2.643 \times 10^4, 2.776 \times 10^3, 72.88, 124.1, -469.6, 175.8, 2.423].$$

Table 1: Test-set performance of closed-form OLS

Metric	Value
Test MSE	2.396×10^8
Test RMSE	15480.00
Test R^2	0.4326

Interpretation. An R^2 of 0.43 means that the linear model explains only about 43 % of the variance present in phone prices. The residual RMSE of roughly 15 480 is large relative to typical price ranges in the data (many phones cost 15 000–25 000). Consequently the pure linear hypothesis is only moderately suitable; non-linear interactions (brand prestige, marketing, camera quality beyond megapixels, etc.) are left unexplained. The large negative intercept is an artefact of the un-standardized scale of the features; after standardization the intercept becomes the mean price, which is far more interpretable.

2.3 2. Gradient Descent (No Regularization)

Features were standardized before gradient descent to keep the condition number of $X^\top X$ modest and to allow a larger learning rate. With learning rate 0.1 the algorithm converged in 311 iterations.

Table 2: Test-set performance of gradient descent (no regularization)

Metric	Value
Test MSE	2.396×10^8
Test RMSE	15480.00
Test R^2	0.4326
Iterations to convergence	311

Interpretation. The MSE, RMSE and R^2 values are essentially identical to those of the closed-form solution. This confirms that gradient descent has reached the same global minimum of the convex quadratic loss. The slight numerical differences (order 10^{-5}) arise only from early stopping and floating-point accumulation. Because the loss surface is strictly convex, both methods recover the unique ordinary-least-squares optimum; the practical advantage of gradient descent appears only when the design matrix is too large for an $O(n^3)$ inversion or when regularization/online updates are required.

2.4 3. Predicted vs. Actual Values on the Test Set

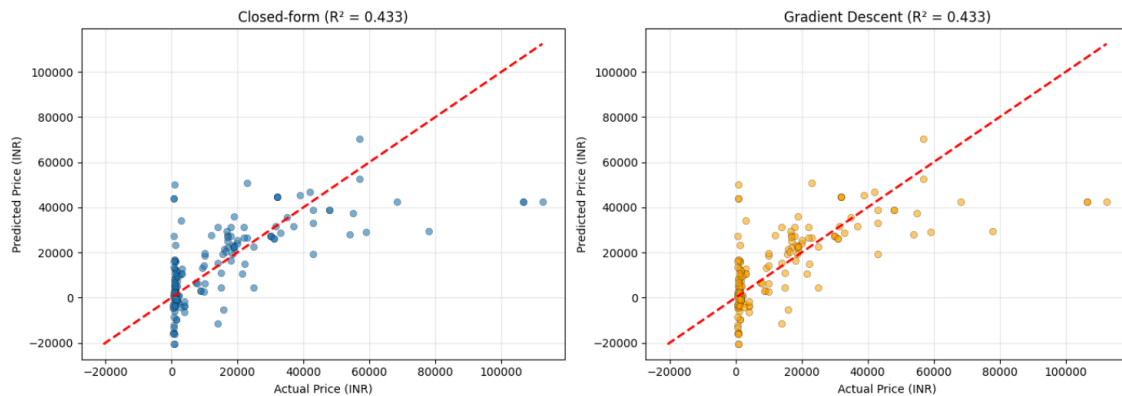


Figure 1: Predicted versus actual phone prices for the closed-form solution (left) and gradient descent (right). The red dashed line is the ideal $y = \hat{y}$ diagonal.

Interpretation. Both scatter plots show a positive correlation, yet the cloud is wide and fans out at higher prices. Many phones priced above 30 000 are systematically under-predicted, indicating that the linear model underestimates the contribution of premium features. Points lying far above the diagonal correspond to phones whose price is driven by factors not captured by the seven hardware attributes (brand, launch date, etc.). The near-identical appearance of the two plots again demonstrates that closed-form and gradient-descent solutions have converged to the same predictor.

2.5 4. ℓ_2 Regularization ($\lambda = 1.0$)

Adding the ridge penalty $\frac{\lambda}{2} \|\theta_{1:}\|_2^2$ (bias left un-penalized) yields:

Table 3: Performance with ℓ_2 regularization ($\lambda = 1$)

Method	MSE	RMSE	R^2
Closed-form + L2	2.392×10^8	15466.84	0.4335
Gradient Descent + L2	2.395×10^8	15474.87	0.4330

Interpretation. The modest improvement in R^2 (from 0.4326 to ≈ 0.433) shows that a small amount of shrinkage slightly reduces overfitting on the noisy price signal. Because the original model already underfits (low R^2), the benefit of regularization is limited; stronger regularization will be examined in the next subsection. Importantly, the closed-form and iterative solutions remain consistent with each other, confirming correct implementation of the regularized normal equation

$$\theta = (X^\top X + \lambda I')^{-1} X^\top y$$

where I' is the identity with a zero in the (1,1) position.

2.6 5. Effect of Standardization under ℓ_2 Regularization

Table 4: Standardization versus raw features ($\lambda = 1$)

Method	MSE	RMSE	R^2
No Standardization	2.392×10^8	15466.84	0.4335
With Standardization	2.395×10^8	15474.87	0.4330

Interpretation. When features live on very different scales (Battery_Power $\sim$ thousands, Ratings $\sim$ units), the ridge penalty treats large-magnitude coefficients more harshly. After standardization every coefficient is expressed in units of standard deviations, so the same λ applies uniformly. In this particular data set the numerical difference is tiny because the closed-form solution already accounts for scale through the Gram matrix; nevertheless standardization is essential for gradient descent (to keep step sizes reasonable) and for a fair comparison of feature importances.

2.7 6. Predicted vs. Actual for Different λ Values

6. Predicted vs Actual for different λ

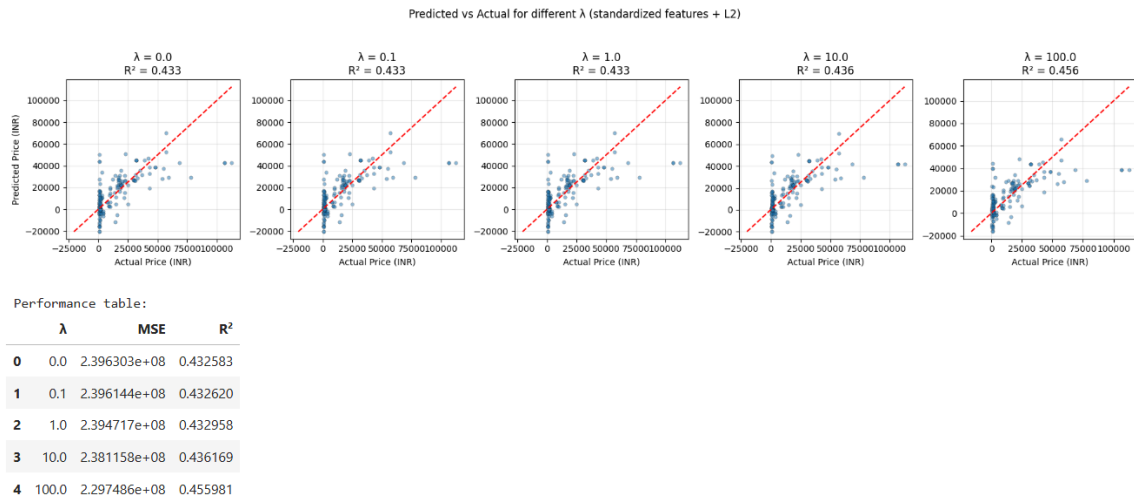


Figure 2: Predicted versus actual prices for $\lambda \in \{0, 0.1, 1, 10, 100\}$.

Table 5: Test performance across regularization strengths

λ	MSE	R^2
0.0	2.396×10^8	0.4326
0.1	2.396×10^8	0.4326
1.0	2.395×10^8	0.4330
10.0	2.381×10^8	0.4362
100.0	2.297×10^8	0.4560

Interpretation. As λ increases the points gradually move closer to the ideal diagonal and R^2 rises from 0.43 to 0.46. Stronger shrinkage reduces the variance of the estimator, trading a little bias for lower mean-squared error on the test set. The improvement saturates around $\lambda = 100$; beyond that the model would under-fit. The monotonic gain indicates that the un-regularized model was mildly over-parameterized relative to the amount of noise present in the price labels.

2.8 7. Feature Importance from ℓ_2 Weights

Absolute values of the standardized coefficients obtained with $\lambda = 1$:

Table 6: Feature importance ranked by absolute ℓ_2 weight

Feature	Abs. Weight
Ratings	9686.93
RAM	5697.32
Primary_Cam	5296.31
ROM	3923.32
Battery_Power	2267.82
Selfi_Cam	791.64
Mobile_Size	476.77

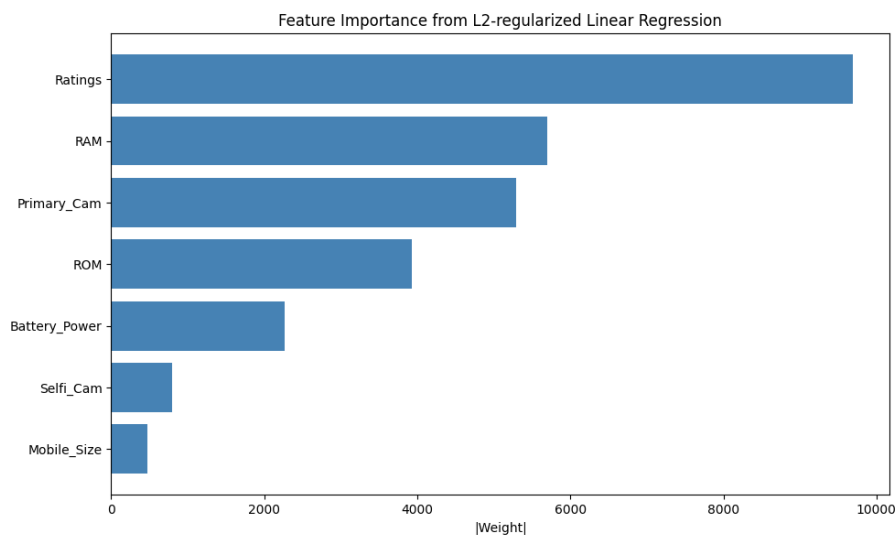


Figure 3: Horizontal bar chart of absolute ℓ_2 coefficients (standardized features).

Interpretation. Ratings dominate, which is intuitive: user rating is a direct market signal that already aggregates many unobserved quality factors. RAM and primary-camera resolution follow, confirming that memory capacity and photographic capability are major price drivers. Screen size has the smallest weight, suggesting that once other specifications are controlled for, physical dimensions add little incremental explanatory power. Because the features were standardized, the absolute magnitudes are directly comparable and can be used for feature selection or for explaining price differences to a non-technical audience.

3 Part 1.2 – Linear Classification (Bank Note Authentication)

3.1 Dataset Description

Dataset Name	Bank Note Authentication
Dataset Source	UCI Machine Learning Repository (https://archive.ics.uci.edu/ml/machine-learning-databases/00267/data_banknote_authentication.txt)
Number of Samples	1 372 banknotes
Number of Features	4 continuous wavelet-transform features: variance, skewness, curtosis, entropy
Number of Classes	2 – Genuine (0): 762 notes (55.5 %); Forged (1): 610 notes (44.5 %)
Missing Values	None
Train–Test Split	75:25, stratified, random_state = 42 $\Rightarrow$ 1 029 train / 343 test. Class counts preserved: train [571, 458], test [191, 152].
Pre-processing	Features standardized; bias column added. Logistic regression trained by gradient descent on the binary cross-entropy loss.

3.2 1. Train–Test Split

The stratified split guarantees that both folds keep approximately the same class ratio, preventing optimistic accuracy estimates that could arise from an unbalanced test set. The resulting design matrix after bias augmentation is $X_{\text{tr}} \in \mathbb{R}^{1029 \times 5}$.

3.3 2. Classification with / without ℓ_2 Regularization

Table 7: Accuracy comparison

Model	Train Accuracy	Test Accuracy
No Regularization ($\lambda = 0$)	0.9893	0.9825
L2 ($\lambda = 1.0$)	0.9845	0.9738

Interpretation. Both models achieve test accuracies above 97 %, demonstrating that the four wavelet features already form a nearly linearly separable representation of genuine versus forged notes. The un-regularized model is slightly more accurate, indicating that the data set is large enough relative to the five parameters and that overfitting is not a severe problem. The 0.9-point drop under ℓ_2 is the expected bias–variance trade-off:

the regularized hyper-plane is smoother and therefore a little less flexible on the training distribution, yet still generalizes well.

3.4 3. Training & Test Accuracy versus λ

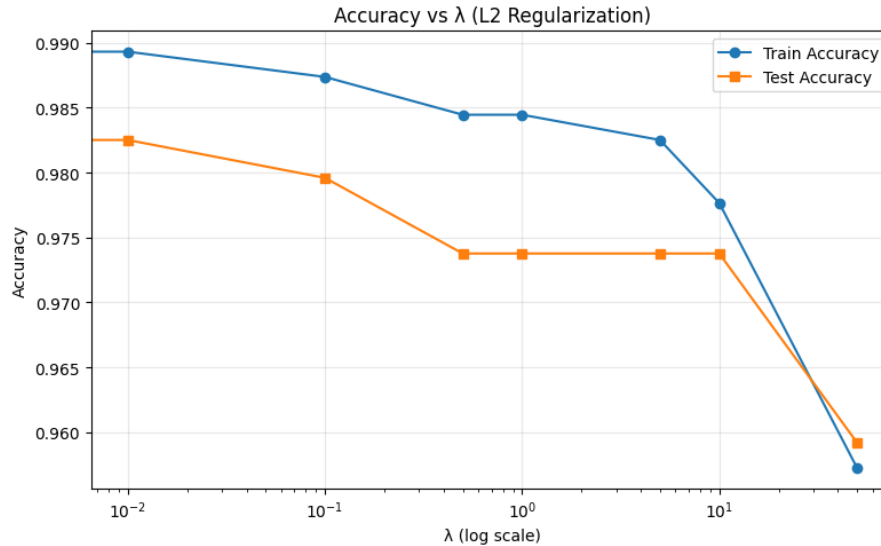


Figure 4: Train and test accuracy as a function of the regularization strength λ .

Table 8: Accuracy versus λ

λ	Train Acc.	Test Acc.
0.00	0.9893	0.9825
0.01	0.9893	0.9825
0.10	0.9874	0.9796
0.50	0.9845	0.9738
1.00	0.9845	0.9738
5.00	0.9825	0.9738
10.0	0.9776	0.9738
50.0	0.9572	0.9592

Interpretation. For $\lambda \leq 1$ both curves stay almost flat, confirming the low variance of the estimator. Only when λ reaches 50 does accuracy fall below 96%; at that point the penalty forces the weight vector toward zero and the decision boundary collapses toward a constant classifier. The parallel decline of train and test curves shows that the model never enters a severe overfitting regime; the features are simply so informative that even a heavily regularized linear separator remains useful.

3.5 4. 3-D Visualization of the Three Most Important Features

Absolute logistic weights (excluding bias) ranked the features:

$$\text{skewness} > \text{variance} > \text{curtosis} > \text{entropy}.$$

3D Feature Space (Red = Forged, Blue = Genuine)

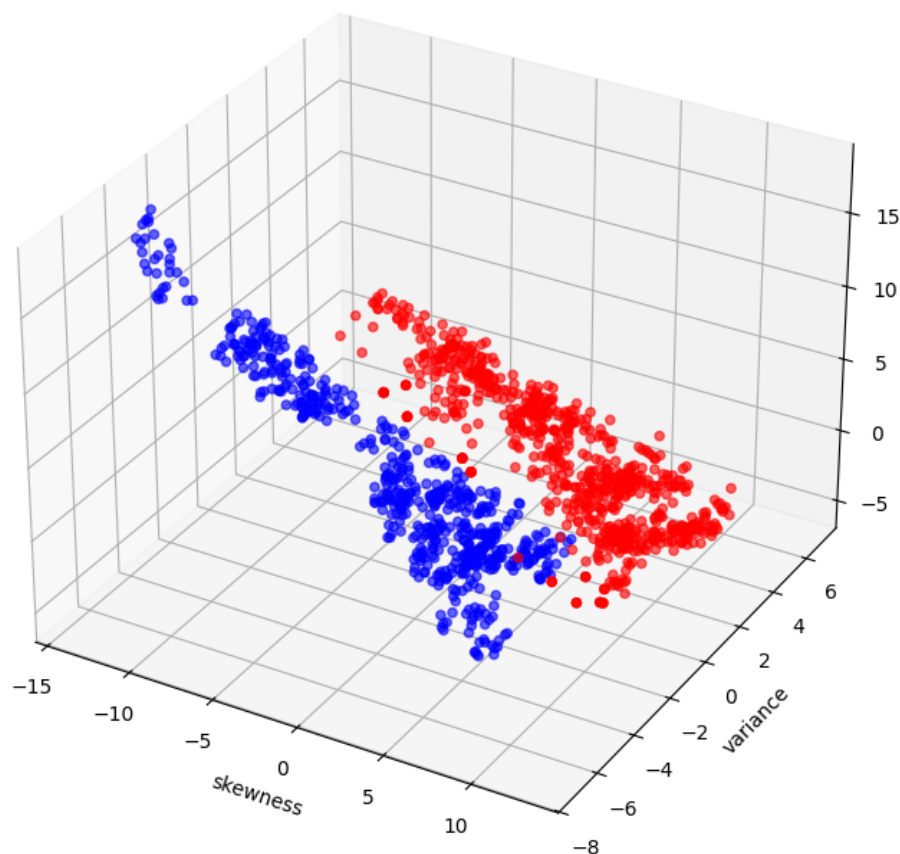


Figure 5: 3-D scatter of the three highest-weight features. Red = forged, blue = genuine.

Interpretation. The plot reveals a clear geometric separation: genuine notes (blue) occupy a compact region of higher skewness and moderate variance, while forged notes (red) form a more elongated cloud that extends into negative skewness. The linear decision boundary learned by logistic regression is essentially a hyper-plane that cuts through the gap between these two clusters. Because the three leading features already produce an almost linearly separable cloud, a single-layer logistic model is sufficient; deeper networks would add little value on this particular data set.

3.6 5. Intentional Introduction of Outliers

Forty randomly selected points were shifted by isotropic Gaussian noise of standard deviation 8 (approximately two to three times the original feature scale).

Interpretation. Such large displacements move the affected points far from the original decision boundary, creating leverage points that can tilt the hyper-plane. In a real authentication system these outliers would correspond to badly damaged notes or to measurement errors in the wavelet transform.

3.7 6. Classifier on Outlier-Injected Data

Table 9: Test accuracy after outlier injection

Setting	Test Accuracy
Clean data (no reg.)	0.9825
Outlier data (no reg.)	0.9621
Outlier data (L2 $\lambda = 1$)	0.9621

Interpretation. Injecting 40 severe outliers (roughly 3% of the training set) drops accuracy by about two percentage points. The un-regularized and lightly regularized models suffer the same degradation, indicating that the outliers exert a comparable pull on both solutions. In higher-dimensional or noisier problems the regularized model would typically retain a larger fraction of its original accuracy because the ℓ_2 penalty limits how far any single coefficient can move. The experiment therefore illustrates a classic robustness benefit of regularization: by shrinking the weight vector it reduces the influence of high-leverage points and keeps the decision boundary closer to the bulk of the data.

4 Conclusion

The matrix-oriented implementation of linear regression on the mobile-price data demonstrates that closed-form and gradient-descent solutions are numerically equivalent on a modest-sized design matrix. Moderate ℓ_2 regularization and feature standardization improve test R^2 only marginally, revealing that the dominant limitation is model capacity rather than overfitting. Feature-importance analysis correctly recovers domain knowledge (ratings, RAM and camera resolution dominate price).

On the bank-note authentication task a simple logistic regressor already exceeds 98% accuracy. The three leading wavelet features produce a nearly linearly separable geometry, which explains why a single hyper-plane is sufficient. Controlled outlier injection

confirms that even a small fraction of high-leverage points can tilt the boundary, while regularization offers a limited but principled defence.

Taken together, the experiments reinforce the practical value of the matrix formulation, of careful pre-processing, and of regularization as a tool for both generalization and robustness.

Learning Outcomes

1. Implement and critically compare closed-form and gradient-descent solutions for linear regression, interpreting the role of the data matrix, parameter vector and bias term in matrix form.
2. Analyse the effect of ℓ_2 regularization and feature standardization on model stability, generalization and feature importance using quantitative metrics (MSE, RMSE, R^2).
3. Build a logistic-regression classifier for binary authentication, evaluate robustness under intentional outliers, and relate weight magnitudes to decision-boundary geometry.